

3D Graph - Maya contest

ANTARCTICA STORY

when Linux and Wireshark get along

TECHNICAL REPORT

SMGv2 animation

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1 Overviews

The movie tells a story of a family penguin in Antarctica. They were spotted by a white shark while resting on a floating iceberg. Although the iceberg was out of reach, the shark made his own approach to the penguins. While everything was under-anticipated according to his plan, the shark encountered something new later on, something that would completely change everything...

2 Pre-production

2.1 Scene list

Because of the complexity in modeling and animation which may require advanced Maya techniques in control and manipulate the objects, the product cannot be filmed in just one simple scene. Instead, to simulate the real filming process, the movie is decomposed into different scenes which has in detail different camera at different perspectives.

For safe developing and testing purpose (as we studied on the way), each scene is constructed several times, each time (as we call it *take*) has some increment in model, technique, effect, etc., until the whole scene is completed. By this way, we can in parallel render each scene (or part of a scene) separately and simultaneously on different computers, which helps improving the efficiency (as we can fix and re-render fault scenes, if any) and thus, reduces time consuming.

The detailed scene list is presented in the Appendix 6.2

3 Modeling

3.1 Penguin model

3.1.1 Construction

The penguin model is built by shaping and combination of basic geometric shapes such as a cone (for the peak) and a sphere (for other body parts). As shown in Figure 1(a), the torso is a sphere scaled along y-axis. The upper half is scaled and moved up while the lower half is scale and moved down. In the similar manner, a wing is a sphere scaled down about along y-axis

and scaled up along x-axis (Figure 1(b)). The peak (Figure 1(c)) is a cone rotated and scaled down along z-axis. A part of the cone is translated inwards, and rotated along x-axis. The eyes are made from 2 simple spheres. A foot is a sphere scaled down along y-axis and having a part translated forward in z-axis (Figure 1(e)). The whole penguin is built by plugging all part together, at the appropriate positions (Figure 1(f)).

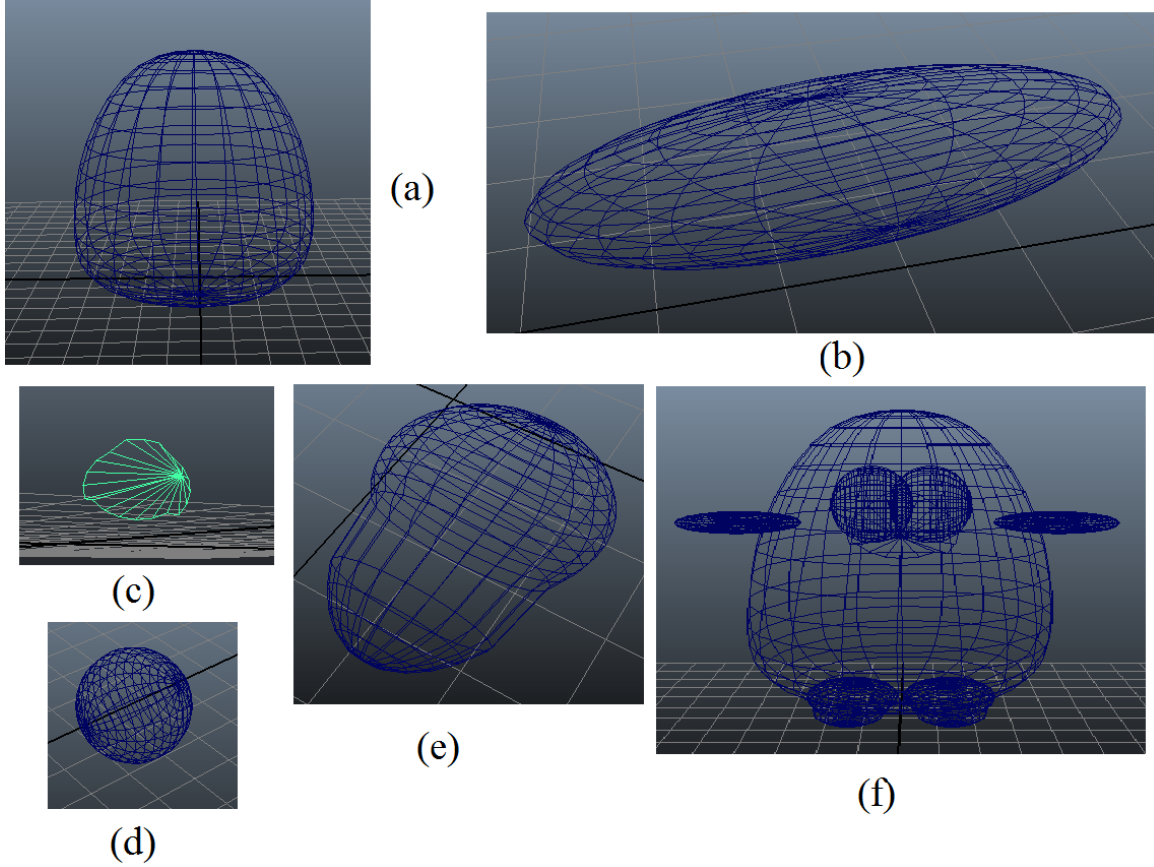


Figure 1: The penguin model. (a) The torso; (b) a wing; (c) the peak; (d) an eye; (e) a foot; (f) all together

3.1.2 Textures

The penguin would not look like one without the textures shown in Figure 2.

3.1.3 Animating skeletons

To animate it, it is necessary to attach a skeleton to the model (Figure 3)

3.1.4 Emotion expression

One of the notable points in this project is that all details from the characters or scenery models to how they animate are constructed from very simple and trivial primitives. The models and animation of a penguin are already shown in the previous parts. However, a simple (or even textured or animated) penguin model would not attract much interest because it behaves just like a robotic statue. To energize these models so that they look lively, we put some emotions onto their faces.

The emotion expression idea is originated from well-known Japanese comics and cartoons, also known as *manga* and *anime*. In these comics and cartoons, we draw characters' emotions

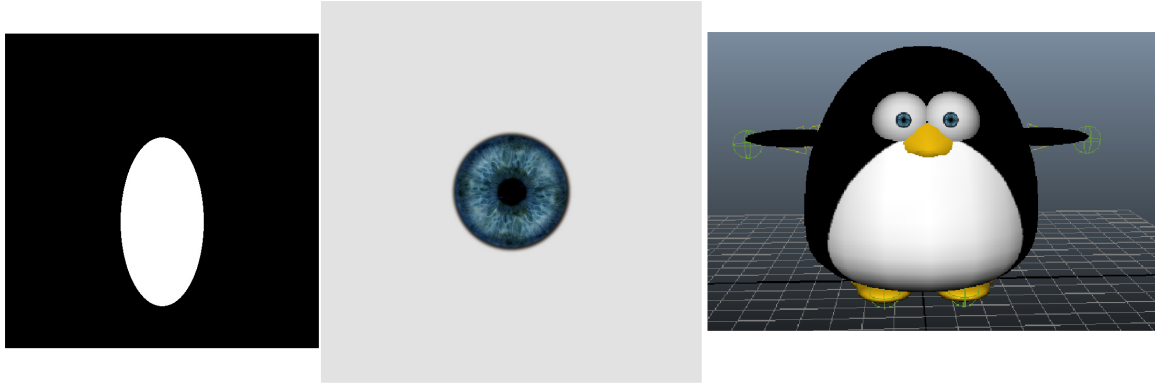


Figure 2: The penguin textures: the body (left), the eye (middle) and the final result (right).

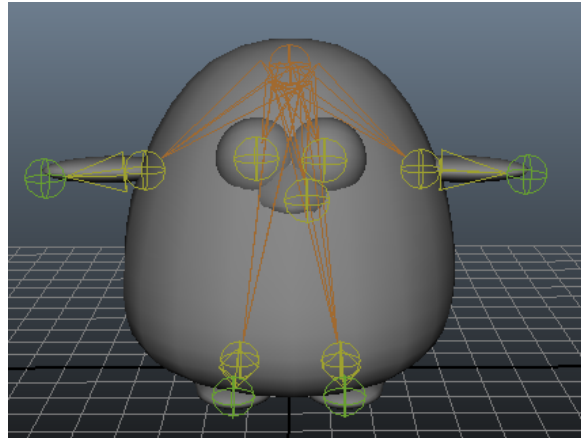


Figure 3: The penguin skeleton

using simple strokes on their faces or on the surrounding environment. As shown on Figure 4, this could be done quite simply by just changing the characters' textures or put some extra ones around the characters. To avoid the side effect caused by environmental lighting, the textures used as emotion expression are all set to transparent, with low diffusion and to reject shadow formation.

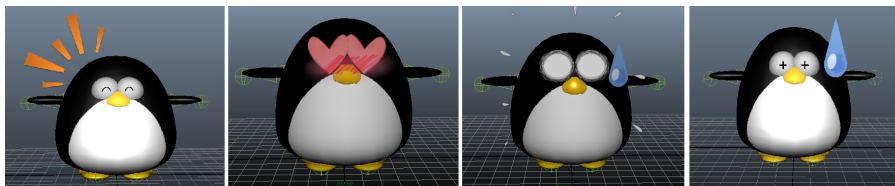


Figure 4: The penguin emotion. From left to right: happy, loved, nervous, anxious

3.2 Shark model

With a bit difference, the shark torso is reconstructed from a downloaded model¹. As shown in the Figure 5

Besides, the shark model is also rigged with skeleton and then textures as the penguin models (Figure 6)

¹<http://www.animationmethods.com/rigs.html>

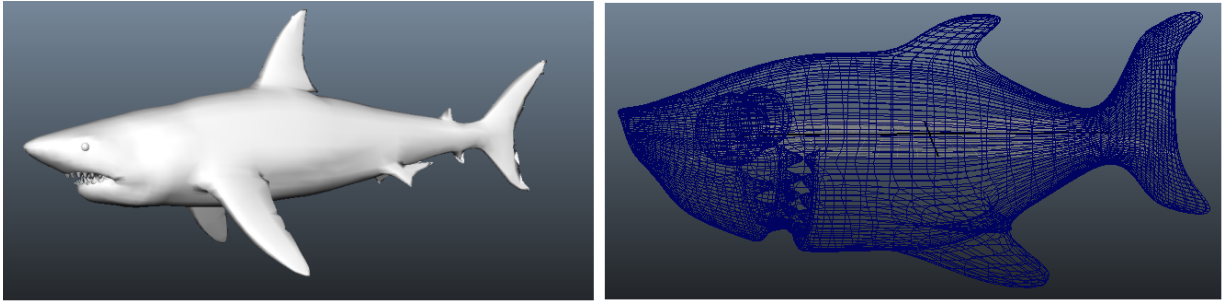


Figure 5: The original shark model (left) and the reconstructed on (right)

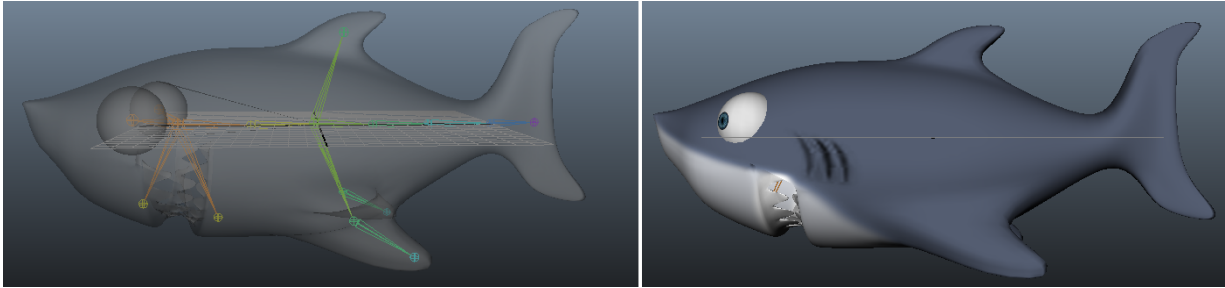


Figure 6: The shark model rigged with skeleton (left) and textures (right)

The emotional expression is created with the same technique as it was to the penguin models: as shown in Figure 7 the emotional strokes is drawn as some extra textures onto the shark

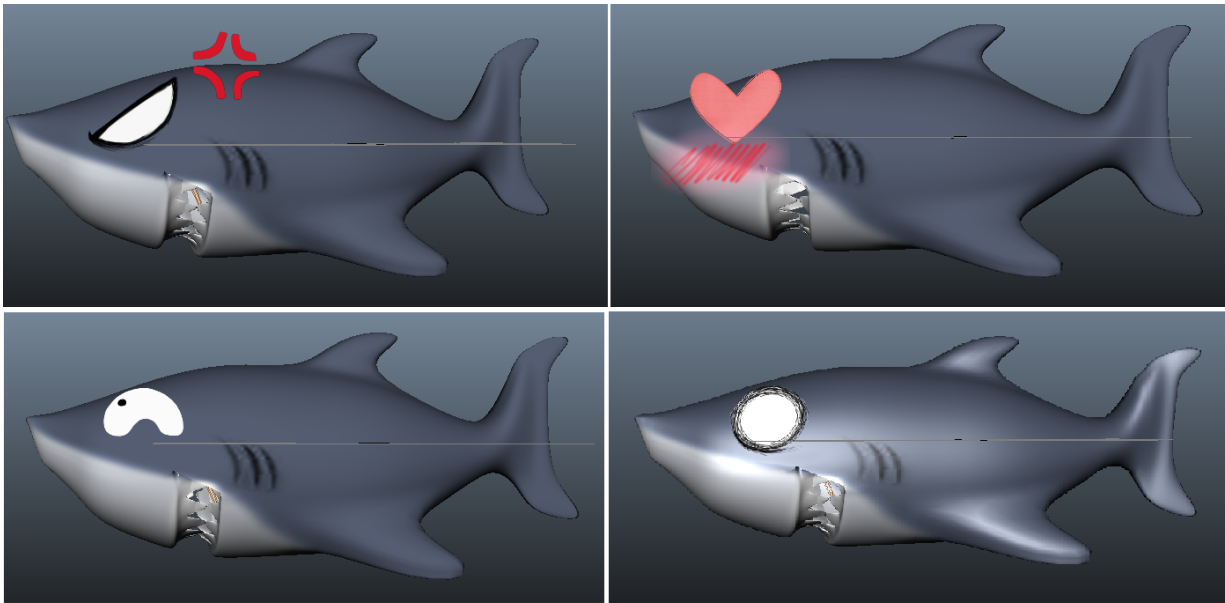


Figure 7: The shark model rigged with skeleton (left) and textures (right)

4 Scenery

4.1 Lanscape

The landscape including the mountains and iceberg is constructed using *sculpt modeling* technique. The technique is motivated from the real sculpting process where a model is developed

from a dense substance such as clay by pull, smooth, grab, pinch or otherwise manipulate each part of the initial mess ². There are 3 types of sculpt modeling including *displacement*, *volumetric* and *dynamic tessellation*. The mountains and iceberg model in this work belong to the displacement type: a dense model, in this case a plane, will have each of its vertex adjusted to a different height (above or below the original plane) which is predefined from a so-called *displacement map* (or *height map*) adjusted locations ³. Figure 8 shows the displacement map used to generate the iceberg. Depending on the parameters set in Maya, we can indicate the height a vertex should be corresponding to a grayscale color provided in the map. Generally, the lighter the color is, the higher a vertex is on a model.

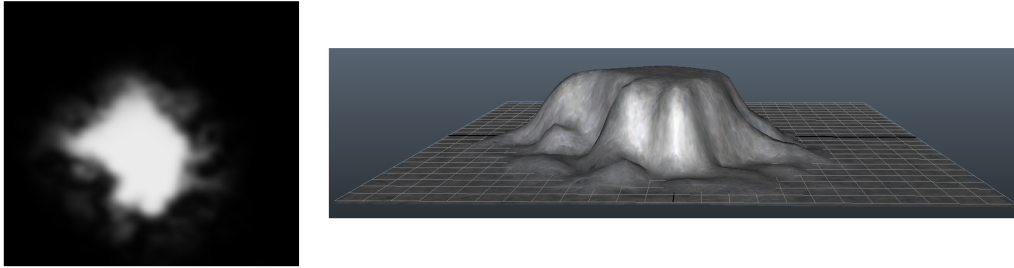


Figure 8: Displacement maps for the iceberg (left) and the result model (right)

Because of the complexities such as the size relation between the landscape size (computed in Maya's unit) and the map size (computed in pixel), the variation in height parameter corresponding with each color on the map, the time consumed (though only about 15 seconds) in generating a model from a map, and the requirement of natural shape of the generated model, construction of the displacement map is really a challenge when working with displacement sculpt model. The landscape used in the production was identified to have a low part in the middle which is later filled with water (where placed the iceberg and starting our story) while has some sort of mountains around the boundary. The mountains should not be plotted as a heap of blocks or some geometric primitive but should be as natural as real mountains. To do so, many displacement maps have been tested (Figure 9), each was generated randomly and after that, retouched with different tools such as blurring, smoothing, blending, etc.

As shown in the left of Figure 10, the samples could not be accepted because they get too artificial with pointed cone shapes while having quite smooth mountainside. Besides, due to uniform color area in a displacement map, the mountains have some staircase-like areas (top-right Figure 10) or unsmoothed surface (bottom-right Figure 10)

4.2 Ocean

The ocean is created using fluid dynamic technique, a built-in function of Maya. However, given the generated ocean as in Figure 12, the challenges are how to make it look like real oceans, including 5 steps

- applying physical sun and sky so that we'll know how the sea would look like in the real lighting environment;
- changing water shading to change the water color according to the sun and sky;
- putting it into scenes to check the compatibility of the ocean color with the mountains' and iceberg's;

²http://en.wikipedia.org/wiki/Digital_sculpting

³http://en.wikipedia.org/wiki/3D_modeling

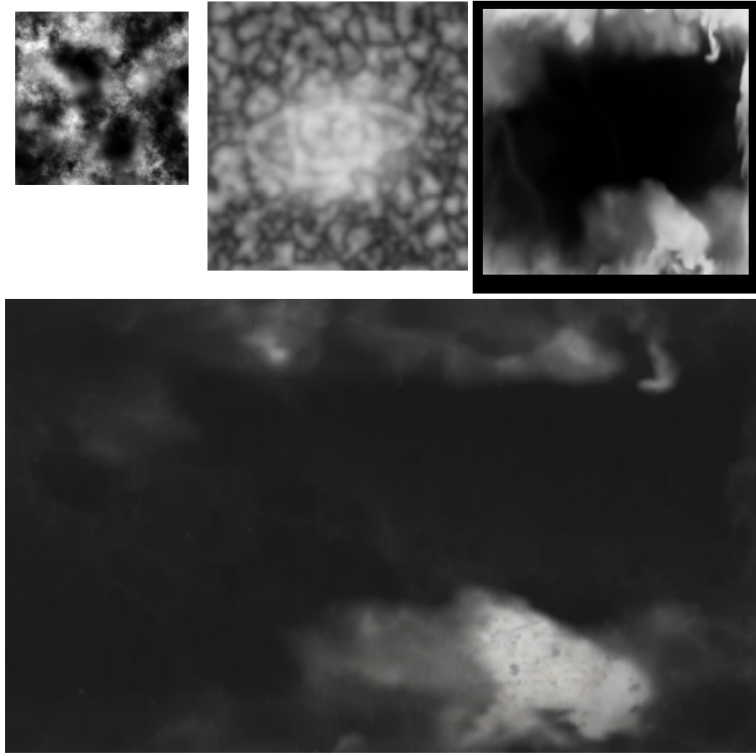


Figure 9: Different tested displacement maps of different size for the landscape. The final accepted one is at the top-right

- and finally, creating wakes, so that the water surface can interact with other objects, making it look more lively.

5 Post-Production

The Maya's rendering technique *mental ray* can only create image files. One of the common format is the Maya's Interchange File Format (or IFF). Thus, to really construct a video out of these images, we need to use another application to do the compositing and more than that, visual effects. The post processing is carried out by using Adobe® After Effect CS6:

Motion graphics Read the Maya's output images and combine them together. This task includes also time controlling: stretch and shrink the duration of each scene so that it fulfills the idea

Sound composition Some effects including trimming, fade in, fade out, amplification, etc.

Visual effects Some of the visual effects used in the clip are

Wiggle effect : to shake a scene to visualize the earthquake or dizzy effect;

Fish-eye effect : makes the scene as if it was seen from the eye of the shark;

Lightning : create some lightning from stormy thunder clouds;

Speed lines : acceleration effect;

Tracing look-path : to show the looking path between 2 objects;

Fireworks : the firework effects are created from an external video.

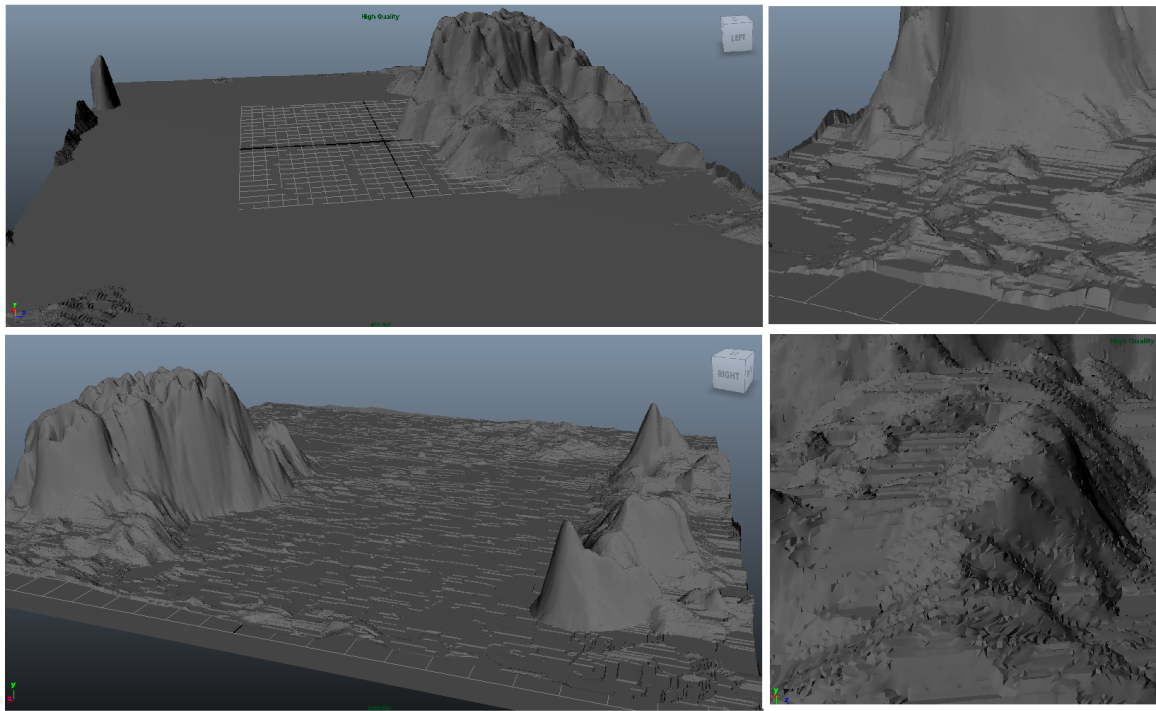


Figure 10: Some sample landscapes (left) and some problems experienced (right)

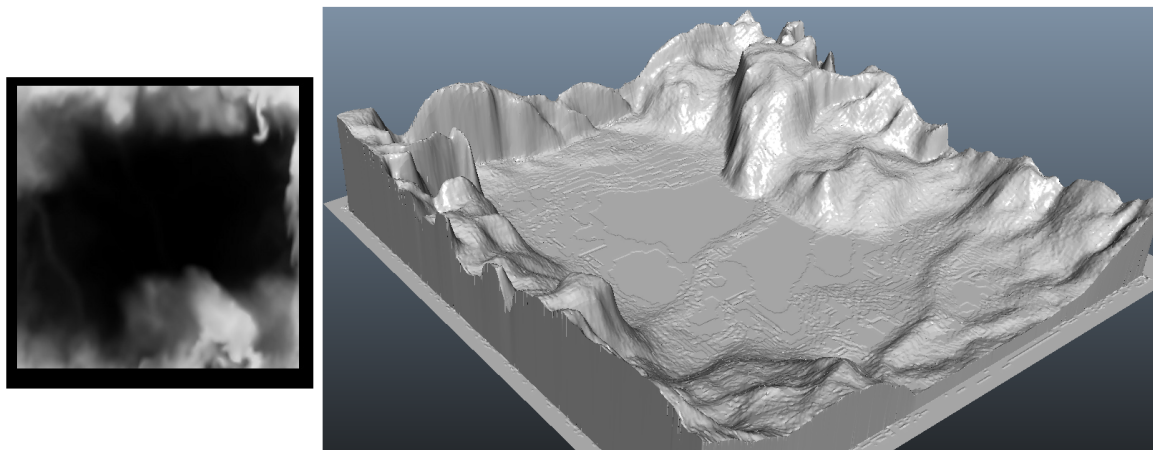


Figure 11: The final landscape version

5.1 Applications

- Autodesk® Maya 2014 - Student version
- Adobe® After Effect CS6
- Adobe® Photoshop CS6
- Adobe® Audition CS6

5.2 Videos

- MGM Logo 3 Roar 2008 Restoration, Youtube
<http://www.youtube.com/watch?v=0VCxJ1aT24A>

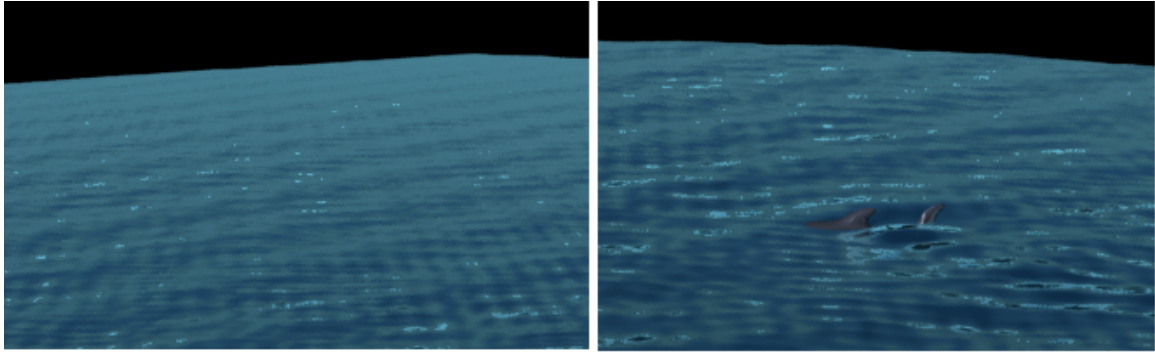


Figure 12: The default water sample generated by Maya (with and without the shark)

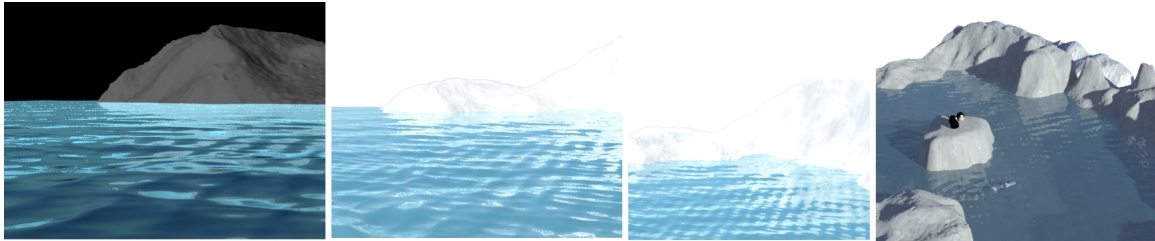


Figure 13: Some failed testing samples (color, lighting effect) and the final version (right-most)

- Miki Owns MGM Lion , Youtube
<http://www.youtube.com/watch?v=LD3JzRZXhs0>
- Muteki Kanban Musume - MGM style, Youtube
<http://www.youtube.com/watch?v=A0DKUq0Vp7g>

5.3 Sounds

5.3.1 Background musics

Opening Sepia No Hi - Card Captor Sakura (Letra), Youtube http://www.youtube.com/watch?v=TeqwDrUK_Ik

Thrilling Let It Go (Disney's "Frozen") Vivaldi's Winter - ThePianoGuys , Youtube
<https://www.youtube.com/watch?v=6Dakd7EIgBE>

Jumping Super Mario Bros Official Theme Song , Youtube
http://www.youtube.com/watch?v=mnipB_8Br8U

Flying Tom and Jerry at MGM music performed live by the John Wilson Orchestra 2013 BBC Proms, Youtube , <http://www.youtube.com/watch?v=viDZza0CXj0>

Ending Tom and Jerry Episode 31 Salt Water Tabby, Youtube
<http://www.youtube.com/watch?v=8IuYFA11nOU>

5.3.2 Sound effects

Evil laugh Tom and Jerry 26 - Solid Serenade Full, Ben McNairy, Youtube
<http://www.youtube.com/watch?v=FQHqR2qmVHM>

Heart thumbing Tom And Jerry 034 - Kitty Foiled (1948), Youtube
<http://www.youtube.com/watch?v=iBEjtpk-9P4>

Falling 2 Tom and Jerry 62 - Cat Napping Full , Ben McNairy, Youtube

<http://www.youtube.com/watch?v=M1Y1aPVQrZ8>

Air Swagging Tom and Jerry 64 - The Duck Doctor Full , Ben McNairy, Youtube

http://www.youtube.com/watch?v=06XQ_wVPM14

Afraid Tom and Jerry 77 - Just Ducky Full, Ben McNairy, Youtube

<http://www.youtube.com/watch?v=T-HyYUmu8J8>

Falling 1 Tom and Jerry 90 - Southbound Duckling Full, Ben McNairy, Youtube

<https://www.youtube.com/watch?v=A0M5sR30VUE>

Surprising Tom and Jerry Cartoon The Flying Cat, Youtube

<http://www.youtube.com/watch?v=eWIUJRxx1QU>

Screaming Tom and Jerry Cartoon Baby Puss, Youtube

<http://www.youtube.com/watch?v=Tlg2bHGxvbc>

Dog barking Tom and Jerry Cartoon That's My Pup!, Youtube

<http://www.youtube.com/watch?v=Wj9izA4D31E>

Kissing Kissing sound effect, Youtube

<http://www.youtube.com/watch?v=Wh4hV1CLaVk>

Crunching sound Person Bites Into Rice Cake Version 2, FreeFx

<http://www.freesfx.co.uk/sfx/bite>

Oh, yeah! Man Saying Oh Yeah, AudioSparx

http://www.audiosparx.com/sa/summary/play.cfm/crumb.1/crumc.0/sound_iid.530621

Thunder Thunder & Lightning Sound Effects [High Quality] , Youtube

<http://www.youtube.com/watch?v=QZpgHrKXooc>

Ocean sound Ocean Seagulls Relaxation, Youtube

<http://www.youtube.com/watch?v=xys0gMPDHFA>

Laughing Jiraiya, The Big Pervert , Youtube

<http://www.youtube.com/watch?v=eBwnVcsJBES>

Popping Super Mario Bros - Jump - Sound Effect [HQ] , Youtube

<http://www.youtube.com/watch?v=9LmPI1NZNUM>

6 Appendix

6.1 Storyboard

6.2 Scene list

Following is the scene list together with the camera details and the computer (at Eurecome) used to render them.

Scene1Take8 (Petzi) : introducing the general scene

Scene2Take25 The shark falls: frame 95 - 155 (camera 3)

Scene2Take4 (Yakari): Camera zoom out, the shark seeing the drumstick appearing on the penguins frame 1 - 85 (camera 1)

Scene2Take5 (Yakari): The shark jumps: frame 85-105 (camera 1 first few frames, camera 2)

Scene3 (hoi Nguyen): scene3.1 : eye moving scene 3.2: happy dancing

Scene4Take2 (Petzi): frame 1 - 60: the shark is angry (camera 4) frame 60 - 150: the shark with the volcano (camera 4)

Scene4Take3 (Petzi): frame 1-95: the shark attacks (camera perspective: coefficient:
Translate: -5.394 3.077 -25.055
Rotate: 8.062 -8.2 0
)

Scene5Take1 (Yakari): frame 1 - 30 (camera 1): still laughing

Scene5Take2 (Yakari): frame 31 - 73 (camera 1): eye bulbing (iceberg falls 1st time)

Scene5Take3 (Yakari): frame 74 - 120 (camera 1) : eye bulbing + sweating (iceberg falls 2nd time)

Scene5Take4 (Yakari): frame 121 - 140 (camera 1): camera zoom out

Scene5Take5 (Yakari): frame 141 - 210 (camera 1): 1st penguin flies

Scene5Take6 (Petzi): frame 211 - 270 (camera 1): 2nd penguin flies

Scene5Take7 (Yakari): frame 271 - 370 (camera 1): 3rd penguin flies + icerberg sunk + shark going forward

Scene6Take1 (Petzi): frame 1 - 100 (camera 1): shark laughs pevertedly

Scene7Take0 (Yakari): frame 371 - 460 (camera 1): 3rd penguin flies up

Scene7Take1 (Yakari): frame 451 - 520 (camera 1): 3rd penguin rotates

Scene7Take2 (Yakari): frame 437 - 450 (camera 1): 3rd penguin gets chill

Scene7Take3 (Yakari): frame 451 - 470 (camera 1): 3rd penguin blinks

Scene7Take4 (Yakari): frame 471 - 476 (camera 1): 3rd penguin falls

Scene7Take5 (Petzi): frame 471 - 516 (camera 2): 3rd penguin falls (fps)

Scene8Take1 (Yakari): frame 471 - 520 (camera 1): blink eyes

Scene8Take2 (Yakari): frame 471 - 981 (camera 1): love eyes (miss from frame 471 - 520)

Scene9Take1 (TinTin): frame 1 - 360 (camera 1): go to the open sea to end

Scene10Take1 (Yakari): frame 1 - 110 (camera 1): 2 penguins fall down + stand up

Scene10Take2 (Yakari): frame 111 - 170 (camera 1): 2 penguins look at each other

Scene10Take3 (Yakari): frame 171 - 210 (camera 1): 2 penguins change